

Enhancing Korean Automated Essay Scoring with Generative Large Language Models

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M.S. Thesis Defense

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[1] Misgna et al., A survey on deep learning-based automated essay scoring and feedback generation, Artificial Intelligence Review, 2024.
Source: National Education Council, Public Survey Results for the 2022 Revised National Curriculum, reported in Asia Economy, June 22, 2021.

Introduction

- Motivation (1/4)

- ❖ Writing assessment is a core form of performance evaluation, but the **high cost, time burden, and inconsistency** of traditional human scoring have increased the need for **Automated Essay Scoring (AES)**.

Traditional Human Scoring



High time burden

Hours per essay × thousands of essays



High cost

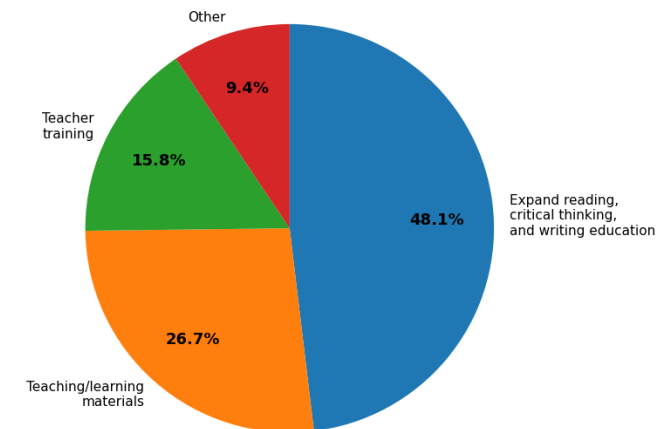
Trained human raters are expensive



Inconsistency

Inter-rater disagreement

Growing Demand for Writing Assessment



<Perceived Support Needs for Essay-Type Assessment>

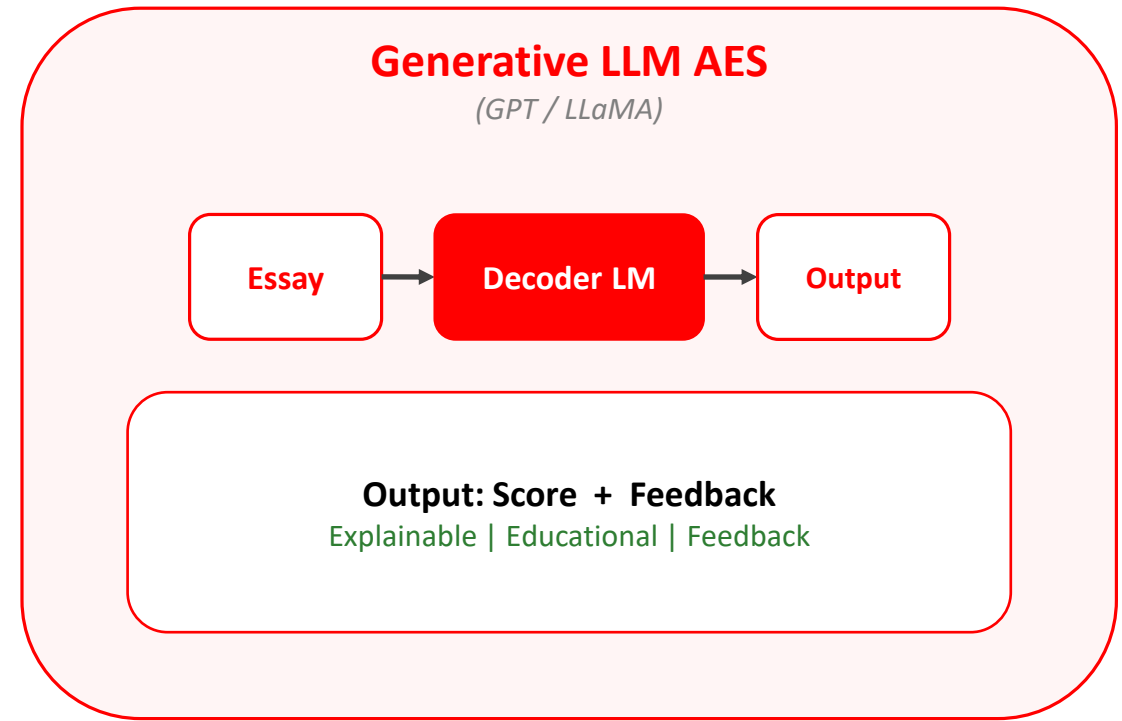
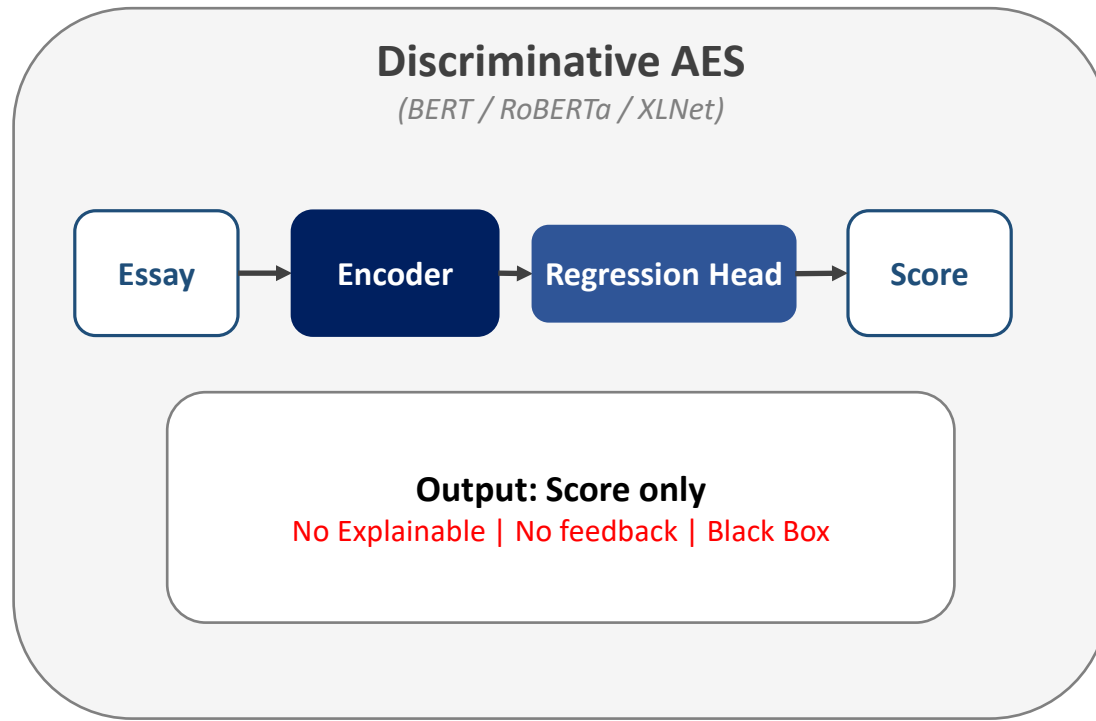
- Nearly half identified expanded reading, critical thinking, and writing education as necessary.

Introduction

- Motivation (2/4)

- [6] Y. Wang, C. Wang, R. Li, and H. Lin, On the use of BERT for automated essay scoring: Joint learning of multi-scale essay representation, NAACL, 2022.
- [7] V. Kumar and D. Boulanger, Explainable automated essay scoring: Deep learning really has pedagogical value, Frontiers in Education, 2020.
- [8] W. A. Mansour, S. Albatarni, S. Eltanbouly, and T. Elsayed, Can large language models automatically score proficiency of written essays?, LREC-COLING, 2024.
- [9] S. Lee, Y. Cai, D. Meng, Z. Wang, and Y. Wu, Unleashing large language models' proficiency in zero-shot essay scoring, Findings of EMNLP, 2024.

- ❖ Discriminative models (BERT/RoBERTa) **score well on Quadratic Weighted Kappa(QWK)** but cannot produce rationale or feedback; generative LLMs unify scoring and feedback in a single pass.



Introduction

- Motivation (3/4)

- [11] J. Zausinger et al., Regress, don't guess: A regression-like loss on number tokens for LMs, ICML 2025.
- [12] S. Ren et al., EMO: Earth Mover Distance Optimization for auto-regressive LM, ICLR 2024.
- [13] S. Yang et al., ThinkAnswer loss: Balancing semantic similarity and exact matching, Findings of EMNLP 2025.
- [15] Y. Song et al., The good, the bad, and the greedy: Evaluation of LLMs should not ignore non-determinism, NAACL 2025.

- ❖ However, Direct scoring with LLM is **not straightforward to optimize for Quadratic Weighted Kappa (QWK)**, because training and inference objectives are misaligned with ordinal scoring.

① Ordinal mismatch

Cross-Entropy treats score tokens as **independent categories**; distance between scores is lost. Predicting 2 when truth is 7 is penalized the same as predicting 6.

② Class imbalance

Real essay datasets cluster on **middle scores**; extreme bands are sparse, so the model collapses toward the mode.

③ Score ↔ feedback bias

Long feedback sequences dominate the token-level loss, so the **score-prediction signal weakens** during training.

④ Stochastic outputs

Sampling-based generation gives **different scores per call** on the same essay — undermining reproducibility.

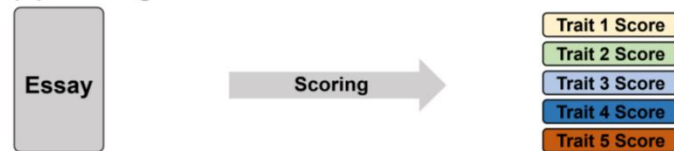
Source: Korean Essay Evaluation with AI (KEEwi), Chu et al. (2025). Rationale Behind Essay Scores: Enhancing S-LLM's Multi-Trait Essay Scoring with Rationale Generated by LLMs. Findings of NAACL 2025.

Introduction

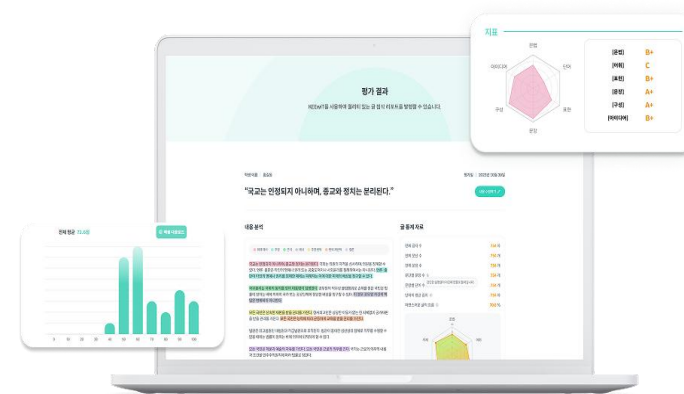
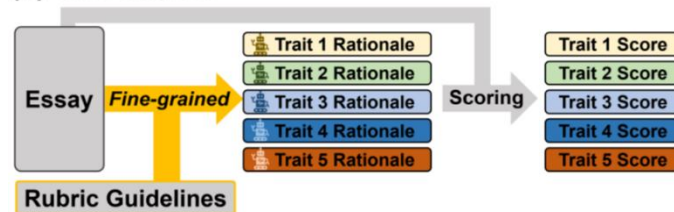
- Motivation (4/4)

- ❖ Recent AES studies use LLMs to generate rationales or feedback that **support separate scoring models**.
- ❖ Practical direction is a **hybrid architecture**, where a discriminative model predicts the score and an LLM generates feedback, as in KEEwi.

(A) Existing Methods



(B) Ours Method



<Rationale Behind Essay Scores (Cai et al., 2025)>

<KEEwi: Korean Essay Evaluation with AI>

- However, these approaches require additional modules and may cause inconsistency between predicted scores and generated feedback.

Introduction

- Goal & Contribution

- ❖ Therefore, this thesis aims to **improve the scoring accuracy** of a single generative LLM-based AES framework by enabling **QWK-oriented training and stable inference**, without changing the backbone architecture.

Related Work

- Evolution of Automated Essay Scoring

[2] E. B. Page, The imminence of ... grading essays by computer, Phi Delta Kappan 47 (5) 1966, 238–243.

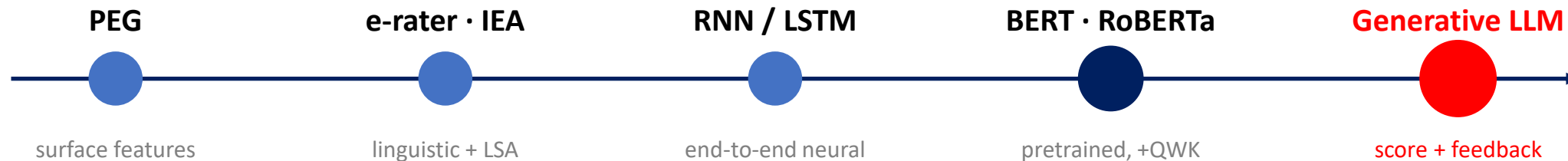
[3] Y. Attali and J. Burstein, "Automated essay scoring with e-rater® V.2," The Journal of Technology, Learning, and Assessment, Vol. 4, No. 3, 2006.

[4] T. K. Landauer, D. Laham, and P. W. Foltz, "The intelligent essay assessor," IEEE Intelligent Systems, Vol. 15, No. 5, pp. 27–31, 2000.

[5] K. Taghipour, H. T. Ng, A neural approach to automated essay scoring, EMNLP 2016, 1882–1891.

[6] Y. Wang et al., On the use of BERT for AES, NAACL 2022

- ❖ AES has shifted from hand-crafted features → neural sequence models → pretrained encoders → generative LLMs.



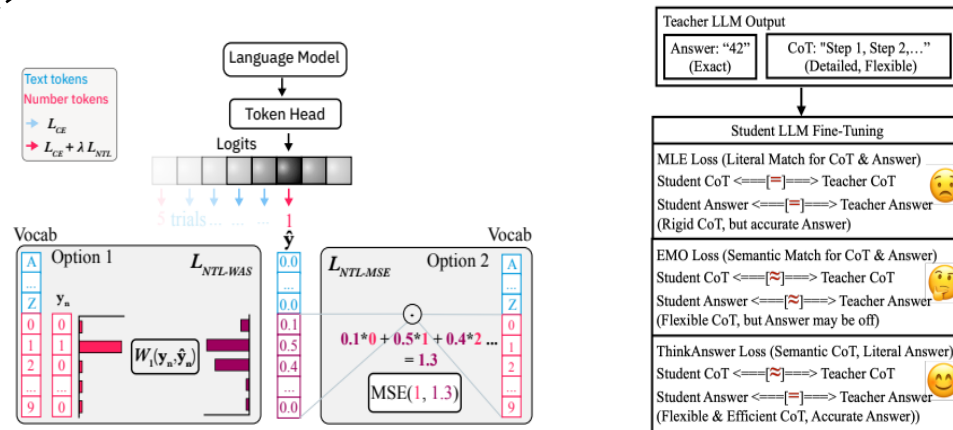
- ❖ AES has improved from feature-based ML to pretrained discriminative models, but this progress introduced an **interpretability trade-off**.
 - ❖ **Feature-based ML** provided limited rationale through **explicit linguistic features**.
 - ❖ **Deep learning** greatly improved QWK but became more **black-box**.
 - ❖ **Generative LLMs** are now explored to generate both **scores and human-readable rationales**.

Related Work

- Generative LLM-based AES and loss-function design

- [10] H. Yoo et al., DREsS: Dataset for rubric-based essay scoring on EFL writing, ACL 2025, 13439–13454.
 [11] J. Zausinger et al., Regress, don't guess: A regression-like loss on number tokens for LMs, ICML 2025.
 [12] S. Ren et al., EMO: Earth Mover Distance Optimization for auto-regressive LM, ICLR 2024.
 [13] S. Yang et al., ThinkAnswer loss: Balancing semantic similarity and exact matching, Findings of EMNLP 2025.

- ❖ Generative LLM AES is **explainable**, but direct LLM scoring often underperforms discriminative models in QWK.
- ❖ Improving **direct scoring within a single generative LLM** framework remains relatively underexplored in AES.



< J. Zausinger et al., Regress, don't guess > < S. Yang et al., ThinkAnswer loss >

- ❖ Building on this idea, this thesis employs these methods for Automated Essay Scoring.

- ❖ Beyond AES, loss design methods have been proposed to better align LLM training with numerical prediction and answer generation.
- ❖ NTL and EMD-based losses[11, 12] address this issue by making **score-token prediction distance-aware and ordinal-aware**.
- ❖ ThinkAnswer Loss[13] motivates decoupled supervision for heterogeneous output tokens.

[14] X. Wang et al., Self-consistency improves chain-of-thought reasoning, ICLR 2023.

[16] K. Zawistowski, Improving LLM comprehension through calculation of expected values, ACSIS 41, 2024.

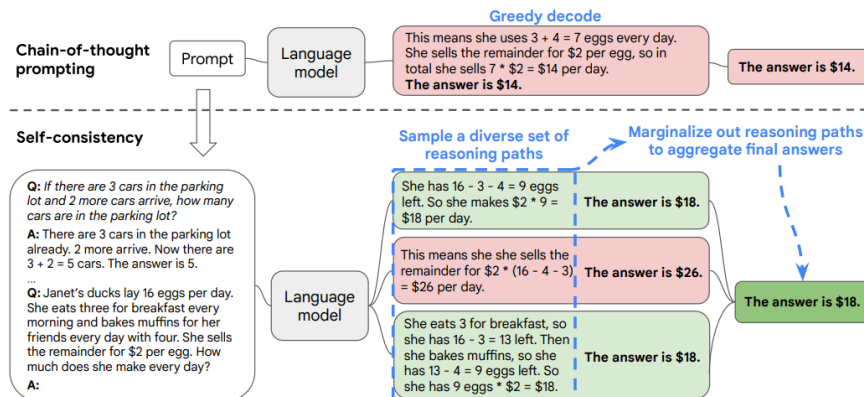
Related Work

- Self-Consistency and Expected-Value Decoding

- ❖ In LLM-based AES, inference strategies are needed to reduce **stochastic score variance** and better utilize **score-token probabilities**.

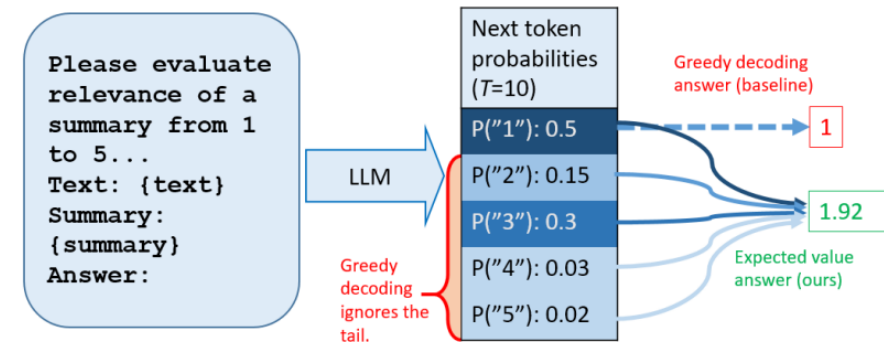
Self-Consistency

Sample m reasoning paths, then aggregate.



Expected-Value Decoding (ED)

Use one forward pass; convert the score-token softmax into an expectation



- ❖ Two inference strategies are applied to stabilize and improve score prediction in generative AES.

Related Work

- Summary

❖ Existing AES research shows a trade-off between **scoring accuracy** and **generative capability**.

Discriminative AES achieves **high QWK**, but **cannot generate feedback**.

LLM-based AES **generates scores and feedback**, but is **difficult to optimize for QWK**.

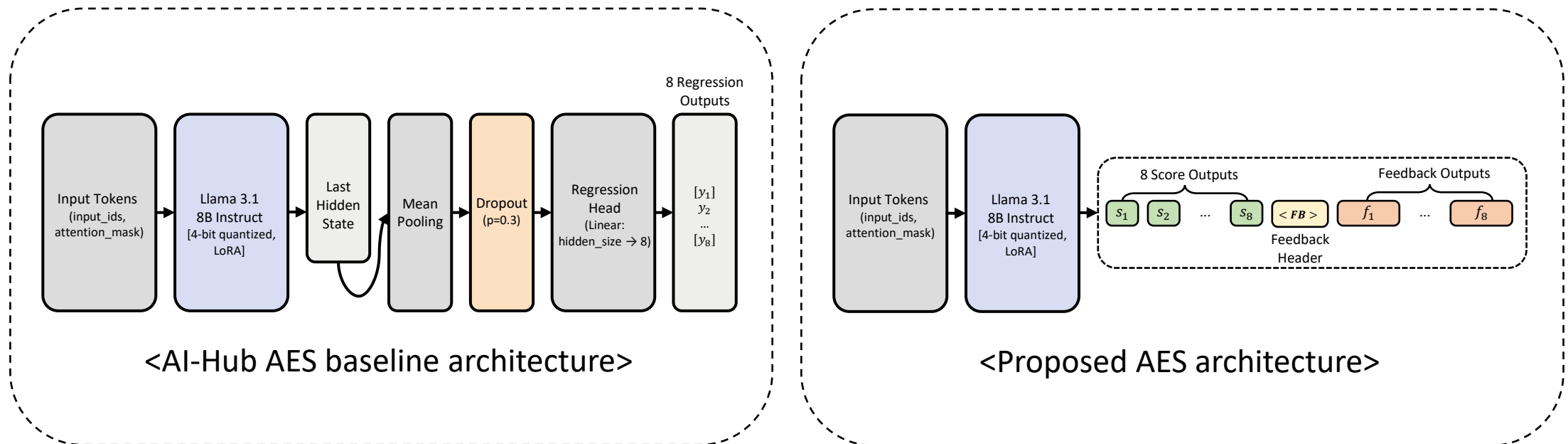
While numerical/probabilistic methods such as **NTL, EMO, ED, and SC** preserve score information, their integration into generative AES frameworks remains underexplored.

This thesis **applies and extends these techniques** to single LLM-based generative AES
-preserving feedback generation.
-improve scoring accuracy

Method

- Preliminary (1/2): From Regression AES to Generative AES

- ❖ We retain the same **Llama 3.1 + LoRA** backbone, but reformulate AES from regression-based scoring to **autoregressive score-feedback generation**.



- ❖ Keep the Llama 3.1 8B backbone fixed and compare regression-head scoring with generative score-token prediction. (8 scores & 8 feedbacks, number of rubric: 8)

Method

- Preliminary (2/2): Why Generative AES Needs Specialized Optimization

- ❖ **Quadratic Weighted Kappa** is a standard AES metric that evaluates ordinal agreement, but it is difficult to optimize directly.

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \quad w_{ij} = (i - j)^2 / (N - 1)^2$$

w_{ij} : quadratic distance weight between predicted i and true j

O_{ij} : observed confusion matrix

E_{ij} : expected matrix under independence

N : number of score classes (here, 9)

$\kappa = 1$: Perfect agreement

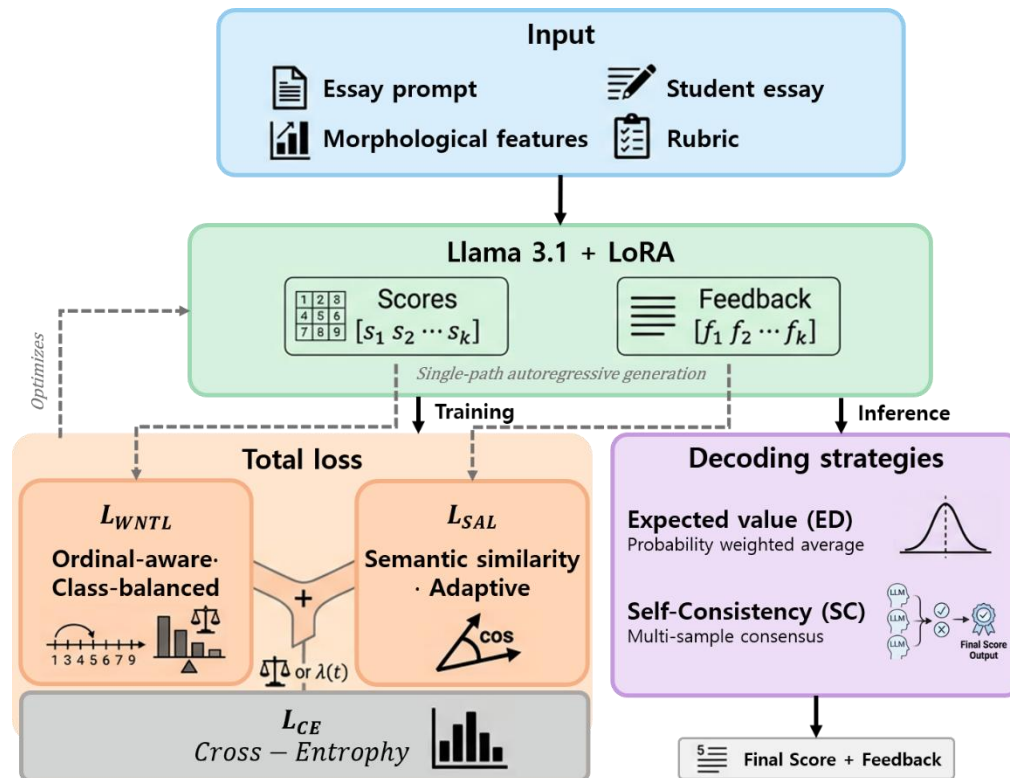
$\kappa = 0$: chance-level agreement

$\kappa < 1$: worse than chance

- ❖ Larger score gaps receive **quadratically larger penalties**.
- ❖ However, QWK is **difficult to optimize** directly because it depends on discrete predictions and corpus-level agreement.
- ❖ Therefore, **MSE-based loss functions** are often used as a practical surrogate, as they naturally penalize larger score differences more strongly and are favorable for optimizing QWK.

Method

- Proposed Algorithm: Overall Process

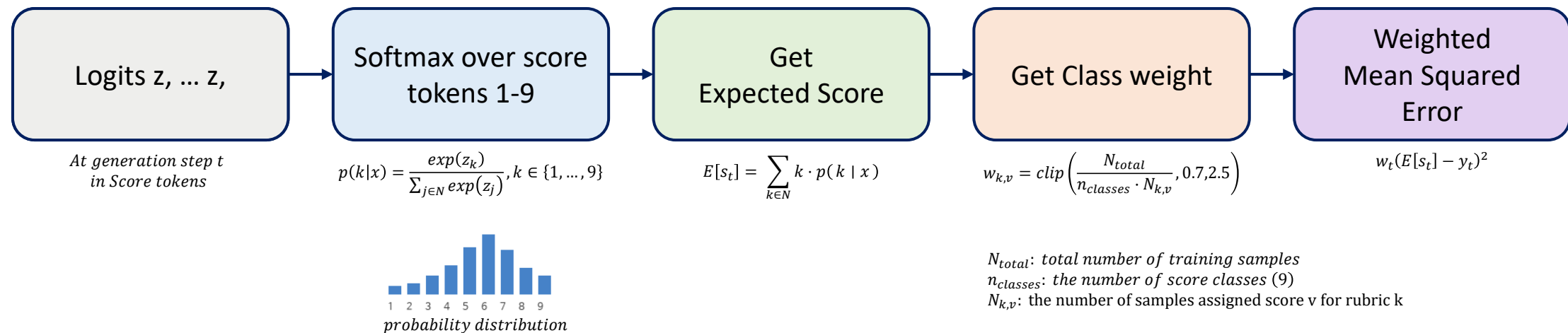


- ❖ Single-path generation
 - ❖ rubric-wise scores and feedback are generated together.
- ❖ **Weighted Number Token Loss (WNTL)**
 - ❖ learns ordinal score relationships and handles class imbalance.
- ❖ **Semantic Alignment Loss (SAL):**
 - ❖ aligns feedback generation semantically and reduces loss imbalance.
- ❖ **Expected Value Decoding (ED) / Self Consistency (SC)**
 - ❖ stabilize score prediction during inference.

Method

- Proposed Algorithm: Weighted Number Token Loss (WNTL)

- ❖ Cross-Entropy ignores the **ordinal distance** between score tokens
 - eg. predicting 2 when the truth is 7 is penalized the same as predicting 6.
- ❖ WNTL pushes score-token probabilities toward the true score and gives more weight to rare scores.

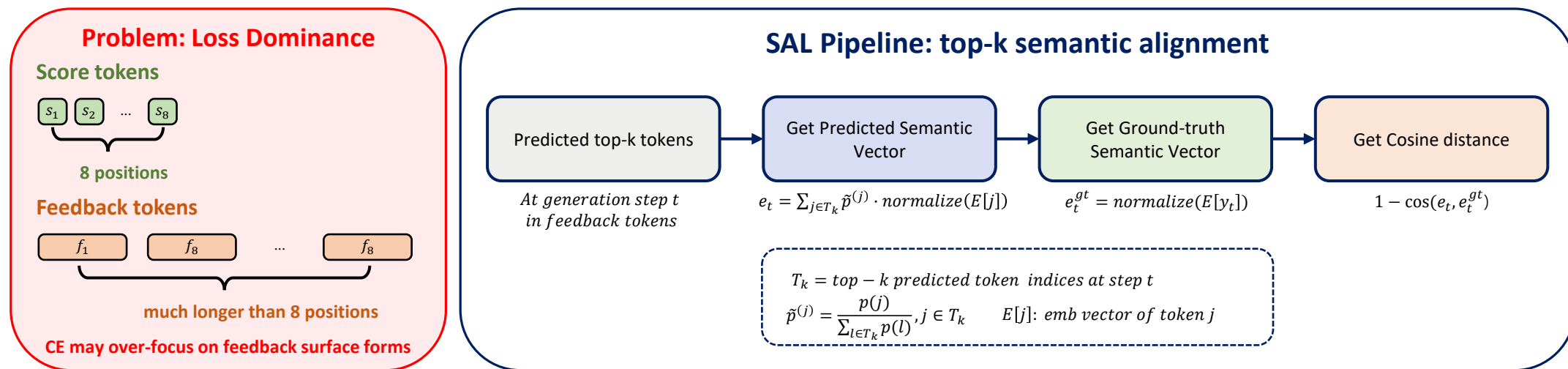


$$L_{WNTL} = \frac{1}{|S_{\text{score}}|} \sum_{t \in S_{\text{score}}} w_t (E[s_t] - y_t)^2, \quad S_{\text{score}} = \text{set of score token position}$$

Method

- Proposed Algorithm: Semantic Alignment Loss (SAL)

- ❖ Feedback tokens are much longer than score tokens, so token-level CE can dominate training
- ❖ SAL gives a semantic supervision signal by aligning predicted top-k token embeddings with gold embeddings



$$L_{SAL} = \frac{1}{|S_{fb}|} \sum_{t \in S_{fb}} (1 - \cos(e_t, e_t^{gt})) , \quad S_{fb} = \text{set of feedback token position}$$

Method

- Proposed Algorithm: Dynamic Loss Balancing

- ❖ CE, WNTL, and SAL have **different numerical scales**; fixed weights can make one objective dominate
- ❖ Dynamic λ **balances CE and auxiliary objectives** at each training step while *detach()* prevents gradients through the weight computation

$$\lambda = \frac{\text{detach}(L_{CE})}{\text{detach}(L_{WNTL} + L_{SAL}) + \varepsilon}$$

$$L_{total} = 0.5 \cdot (L_{CE} + \lambda \cdot (L_{WNTL} + L_{SAL}))$$

$$(scale = L_{CE})$$

L_{CE} : standard cross-entropy on the **full output sequence**

L_{WNTL} : weighted number-token loss on **score tokens**

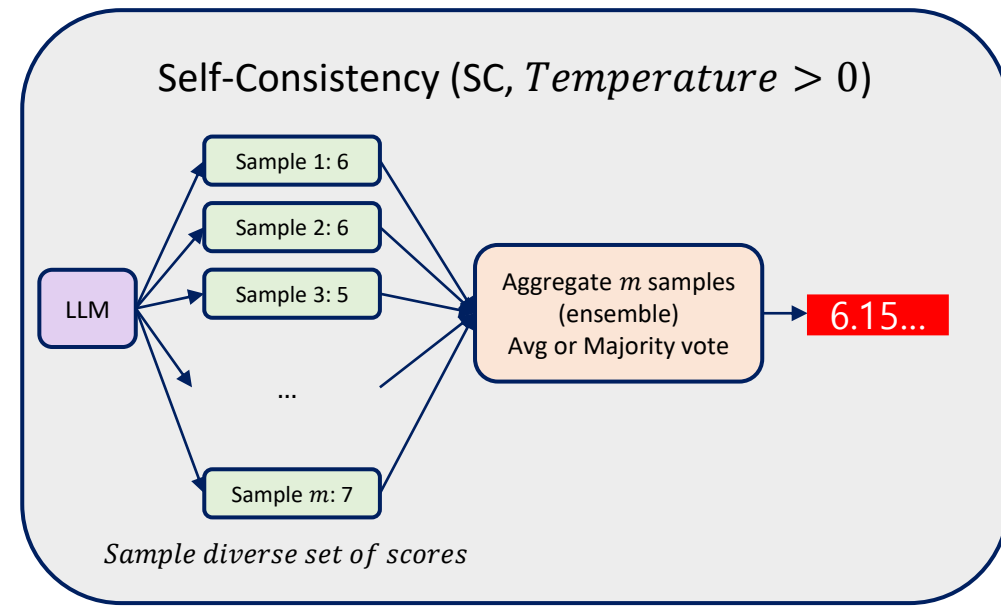
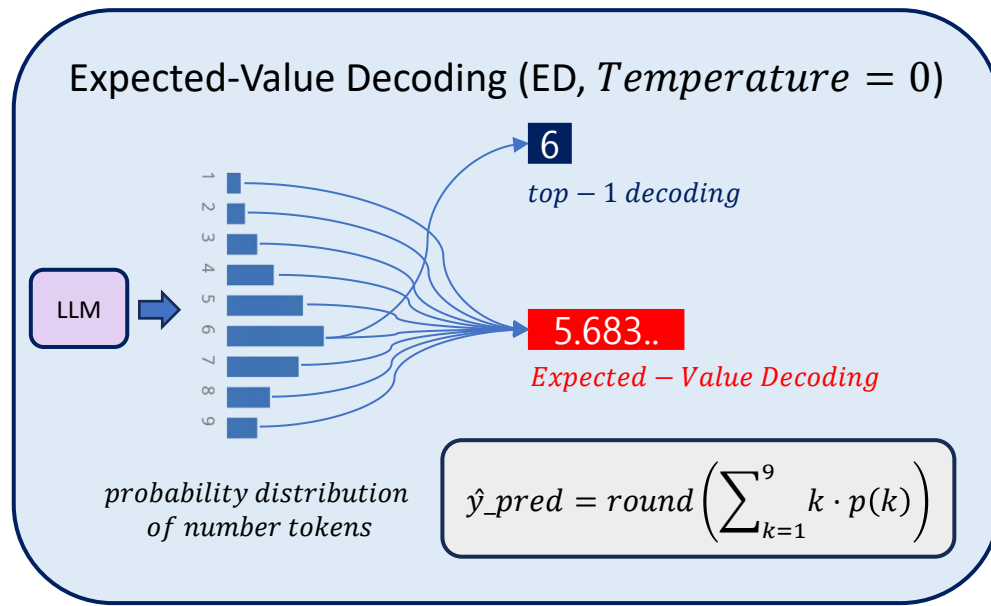
L_{SAL} : semantic alignment loss on **feedback tokens**

ε : **small constant** for numerical stability (10^{-8})

Method

- Proposed Algorithm: Inference Strategies (ED & SC)

- ❖ After training, a generative model still yields **different scores** per call on the same essay — inference-side stabilization is therefore necessary.
- ❖ Two complementary strategies are applied: **Expected-Value Decoding (ED)** for a single-pass ordinal-aware score, and **Self-Consistency (SC)** for multi-sample consensus.



Results

- Dataset and Rubrics

❖ Dataset

- I. AI-Hub Korean essay scoring corpus
- II. Dataset Split: train 32,006 / validation 4,000 / test 4,000
- III. Score Labels: two raters' 1~5 scores converted into 1~9 scale (rater 1+rater 2-1)
- IV. Feedback Labels: raters' rubric-specific feedback descriptions
- V. Target: 8 rubric-wise scores + feedback generation

❖ Rubric

Category	Rubric	Description
Task	과제 수행의 충실성	지시문 요구사항과 조건의 충실한 반영
Cont 1-3	설명의 명료성	설명 대상에 초점을 둔 명확한 서술
	설명의 구체성	구체적 세부 내용과 타당한 근거의 충분한 제시
	설명의 적절성	지시문 요구에 부합하는 내용 구성(핵심어 4개 이상 포함)
Org 1-2	문장의 연결성	문법적으로 정확한 문장 구성과 자연스러운 문장 간 흐름
	글의 통일성	일관된 주제와 흐름의 유지
Exp 1-2	어휘의 적절성	적절하고 다양한 어휘 선택과 자연스러운 문장 표현
	어법의 적절성	문법, 맞춤법, 띄어쓰기의 전반적 정확성

Results

- Experimental Setup

- ❖ All variants share the **same Llama 3.1-8B-Instruct backbone**, identical LoRA / QLoRA configuration, and the same training schedule in A100 GPU.
- ❖ The only differences are the **loss function** and the **inference strategy**, isolating their contribution to the QWK gain.

Training

Backbone Model	Llama 3.1-8B-Instruct
Max Sequence length	2,048
Batch size (per device)	4
Gradient accumulate steps	8
Effective batch size	32
Learning rate	3×10^{-4}
Epochs	10 (early stopping patience = 3)
Precision	BF16
Quantization	QLoRA (NF4, double quantization)
Optimizer	AdamW
Seed	42

LoRA

Rank (r)	16
Alpha (α)	32
Dropout	0.05
Target modules	q-proj, v-proj
Bias	None

Inference (Self-Consistency)

Temperature	0.7
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Results

- Compared Models (ablation)

- ❖ Six configurations are compared, isolating the **contribution of each loss component** from the regression baseline up to the full proposed objective.
- ❖ The final proposed model combines **CE + WNTL + SAL** with dynamic balancing.

Model	Reg. Head	CE	NTL	SAL	Gen.	Description
Baseline	●	—	—	—	—	Llama + Regression Head
CE	—	●	—	—	●	Only CE
NTL	—	●	●	—	●	CE + NTL
NTL+SAL	—	●	●	●	●	CE + NTL + SAL
WNTL	—	●	● _w	—	●	CE + Weighted NTL
SAL+WNTL	—	●	● _w	●	●	CE + SAL + Weighted NTL (Proposed)

Results

- Overall QWK Performance

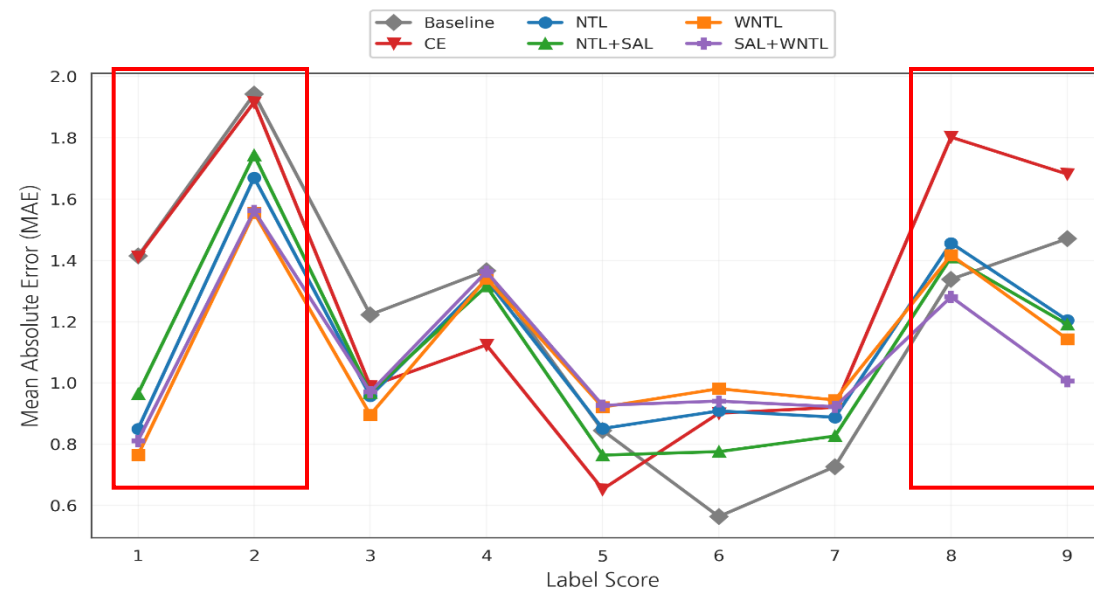
- ❖ CE alone falls **below the regression baseline** — generative structure by itself is insufficient.
- ❖ Adding WNTL and SAL progressively recovers and surpasses the baseline; the full **SAL + WNTL configuration** attains the best Average and Overall QWK.

범주	평가 항목	Baseline	CE	NTL	NTL+SAL	WNTL	SAL+WNTL
Task	과제 수행의 충실성	0.6153	0.5876	0.6300	0.6398	0.6458	0.6521
Cont 1	설명의 명료성	0.5441	0.4732	0.5339	0.5675	0.5638	0.5757
Cont 2	설명의 구체성	0.4867	0.3670	0.4651	0.4931	0.4989	0.5248
Cont 3	설명의 적절성	0.9457	0.9419	0.9500	0.9540	0.9523	0.9548
Org 1	문장의 연결성	0.3051	0.1971	0.3466	0.3516	0.3582	0.3870
Org 2	글의 통일성	0.5905	0.5614	0.5936	0.6064	0.6134	0.6208
Exp 1	어휘의 적절성	0.3271	0.2683	0.3410	0.3952	0.3657	0.3986
Exp 2	어법의 적절성	0.7093	0.6361	0.7800	0.7730	0.7667	0.7798
Average		0.5655	0.5041	0.5800	0.5976	0.5956	0.6117
Overall		0.7049	0.6471	0.6988	0.7178	0.7010	0.7151

Results

- Error by Label Score

- ❖ CE produces **large absolute errors at the extreme bands**, consistent with mode-collapse toward the middle scores.
- ❖ WNTL reduces **ordinal-distance errors**.
- ❖ The proposed loss(SAL+WNTL) flattens the per-label error curve, indicating **better calibration across low, middle, and high scores**.



Results

- Inference Strategy Comparison

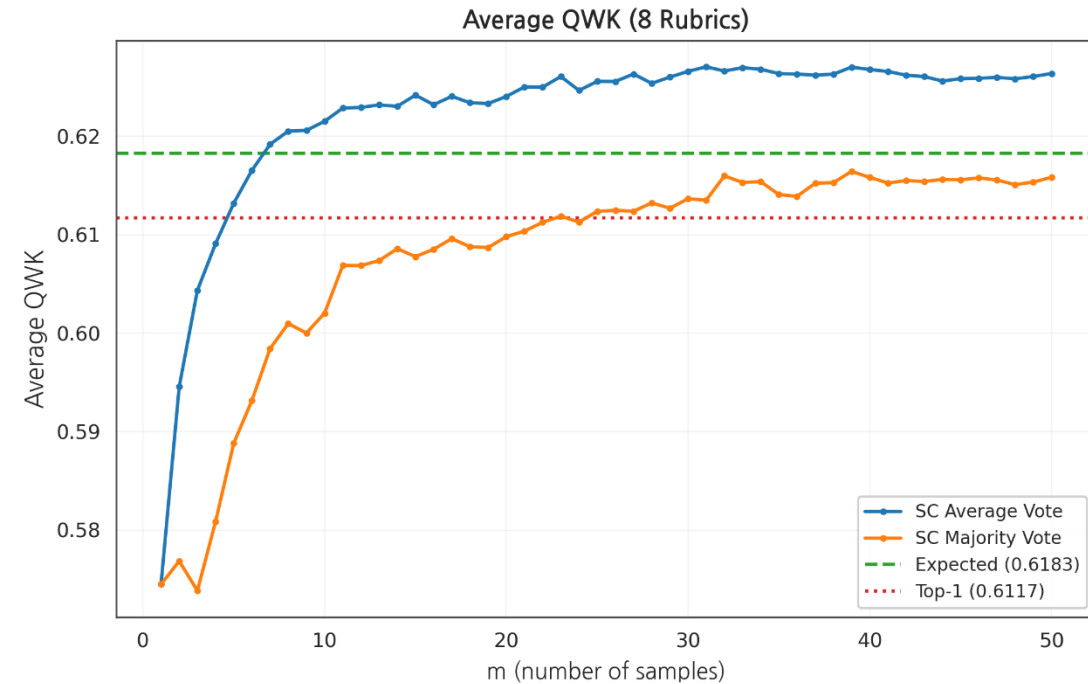
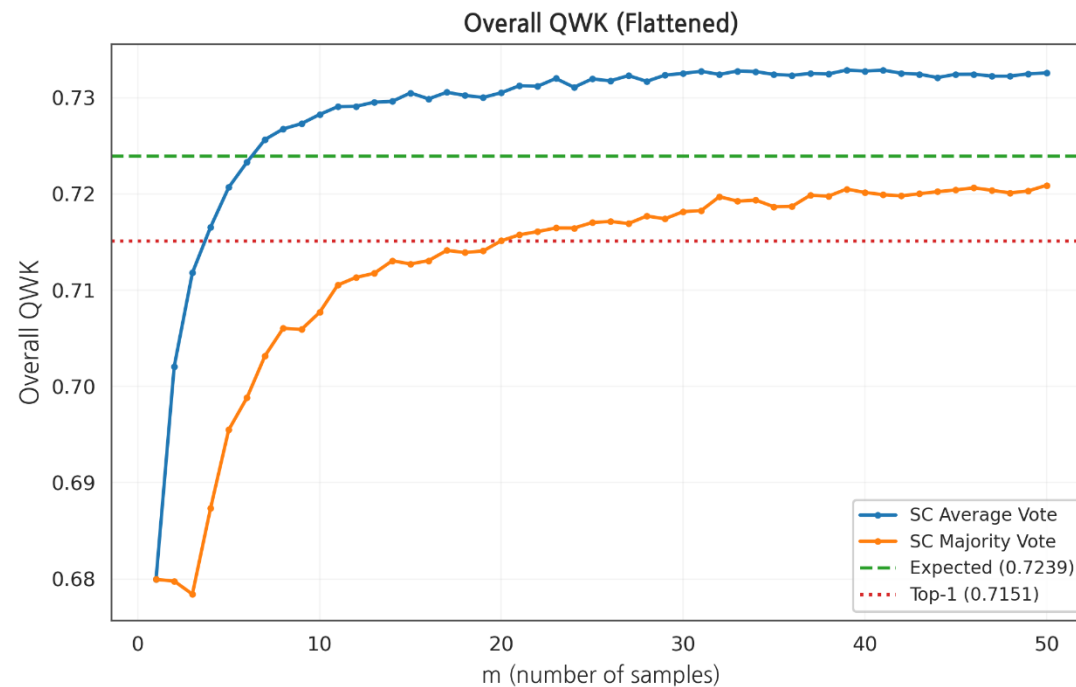
- ❖ **ED** improves over Top-1 without additional generation cost by re-using the score-token probability distribution.
- ❖ **SC-Average yields** the highest Average and Overall QWK; averaging preserves ordinal information, while majority vote discards it.

범주	평가 항목	Top-1	ED	SC-Avg	SC-Maj
Task	과제 수행의 충실성	0.6521	0.6479	0.6521	0.6489
Cont 1	설명의 명료성	0.5757	0.5844	0.5950	0.5816
Cont 2	설명의 구체성	0.5248	0.5400	0.5496	0.5359
Cont 3	설명의 적절성	0.9548	0.9596	0.9597	0.9550
Org 1	문장의 연결성	0.3870	0.3897	0.4098	0.3949
Org 2	글의 통일성	0.6208	0.6216	0.6223	0.6215
Exp 1	어휘의 적절성	0.3986	0.4123	0.4311	0.4118
Exp 2	어법의 적절성	0.7798	0.7907	0.7969	0.7819
Average		0.6117	0.6183	0.6271	0.6164
Overall		0.7151	0.7239	0.7328	0.7205

Results

- Number of Self-Consistency Samples

- ❖ QWK rises rapidly with early samples and **saturates around $m \approx 30$** — diminishing returns past this point.
- ❖ **SC-Average consistently outperforms SC-Majority**; the practical choice is $m = 31$ for an accuracy–cost balance.



Conclusion

- Unified Generative AES for Korean Essay Scoring

- ❖ The framework reaches the best QWK of this study while still producing free-form feedback in the **same generative pass**.

Average QWK	Overall QWK
0.6271	0.7328
+10.9% over the regression baseline	Best result · SC-Average, $m = 31$

<Score + Feedback in one generative pass — without changing the backbone architecture>

Conclusion

- Future Work

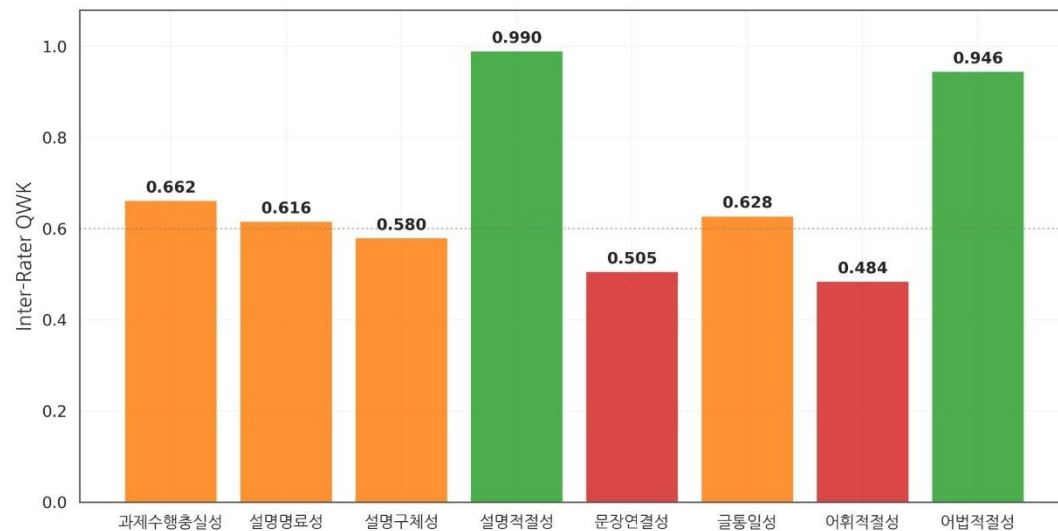
- ❖ Feedback quality evaluation.
 - ❖ Human evaluation and automatic metrics such as BERTScore.
 - ❖ usefulness, and pedagogical validity.
- ❖ Larger backbone and LoRA configuration
 - ❖ Analysis of how larger LLMs, such as 70B-scale models,
 - ❖ broader LoRA adaptation affect scoring performance.
- ❖ Generalization to other writing tasks
 - ❖ Validation on writing genres beyond Korean explanatory essays.
- ❖ Efficient Self-Consistency
 - ❖ Reducing the m-fold inference cost of SC for real-time AES deployment.

**Thank you for
your attention!**

Appendix

- Inter-Rater Agreement

- ❖ Inter-rater QWK indicates how consistently human raters agree on each rubric. **Low agreement** implies intrinsically ambiguous supervision
- ❖ *Average $\kappa > 0.6$*

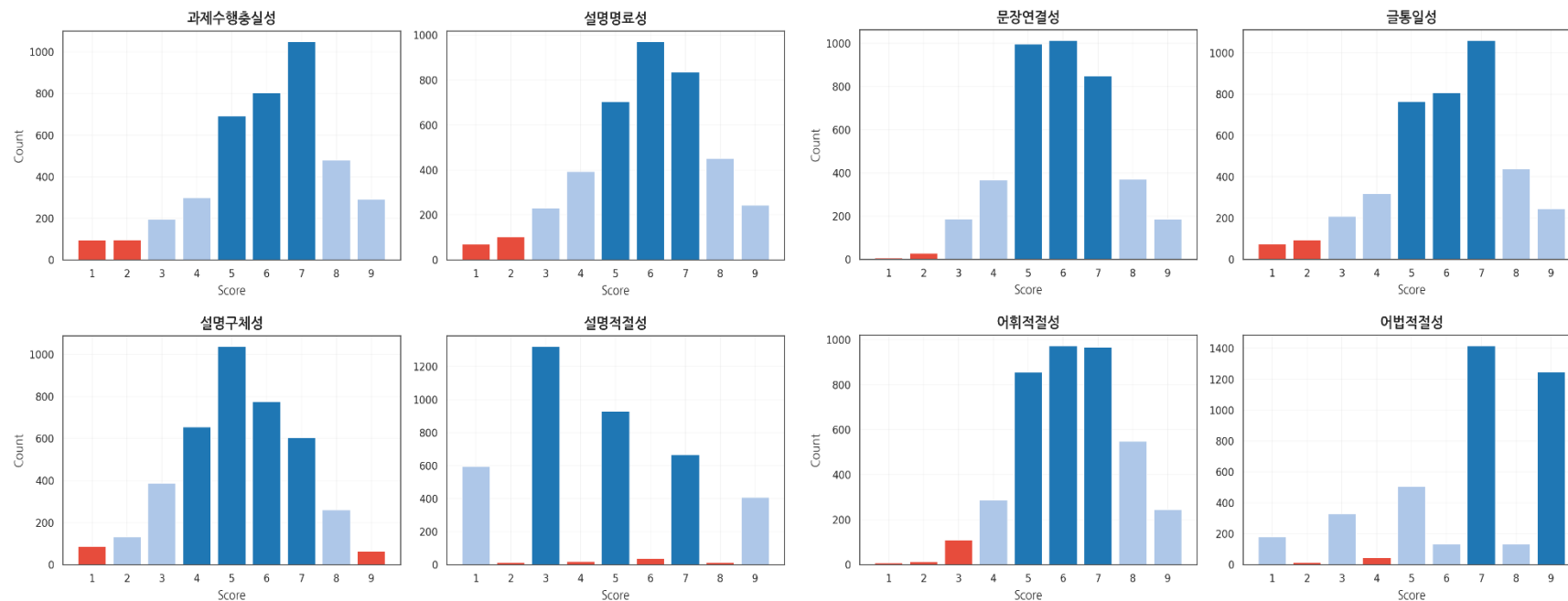


Rubric	Description
과제 수행의 충실성	지시문 요구사항과 조건의 충실한 반영
설명의 명료성	설명 대상에 초점을 둔 명확한 서술
설명의 구체성	구체적 세부 내용과 타당한 근거의 충분한 제시
설명의 적절성	지시문 요구에 부합하는 내용 구성(핵심어 4개 이상 포함)
문장의 연결성	문법적으로 정확한 문장 구성과 자연스러운 문장 간 흐름
글의 통일성	일관된 주제와 흐름의 유지
어휘의 적절성	적절하고 다양한 어휘 선택과 자연스러운 문장 표현
어법의 적절성	문법, 맞춤법, 띄어쓰기의 전반적 정확성

Appendix

- Label Distribution and Class Imbalance

- ❖ Across most rubrics, labels concentrate around the **middle bands (5–7)**, while **extreme bands (1–2, 8–9)** are markedly sparse.



Appendix

- QWK Improvement over the CE Baseline

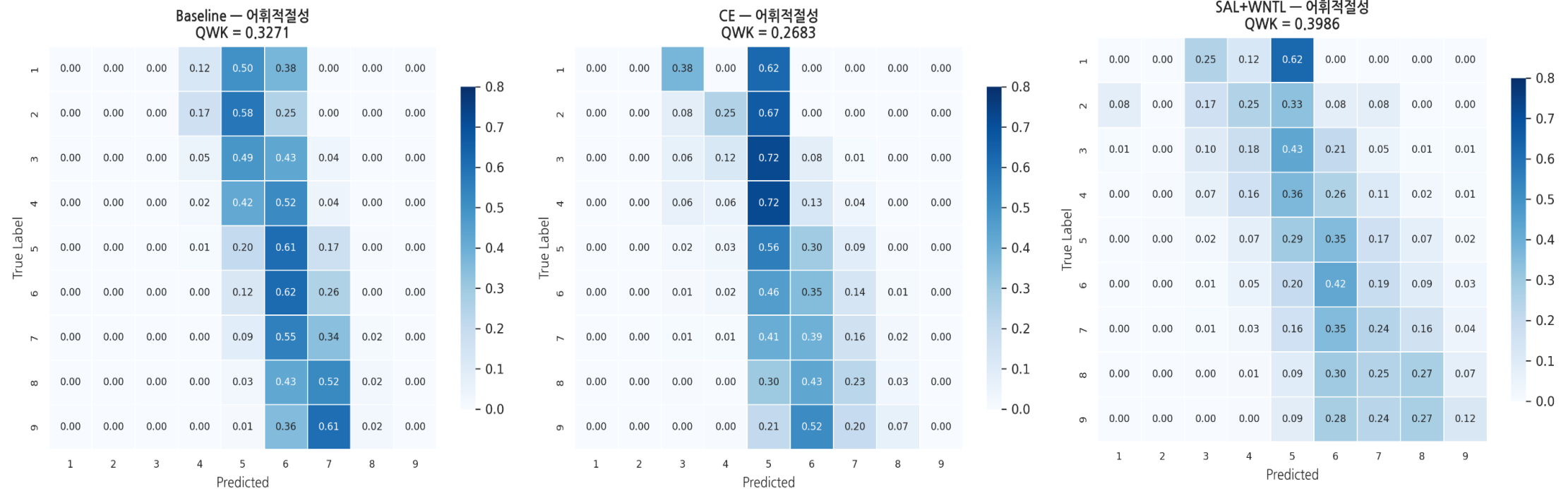
- ❖ Each auxiliary loss yields a positive marginal contribution; their **combination produces the largest overall gain**.
- ❖ The strongest improvements appear on **sentence connectivity** and **lexical appropriateness** — the rubrics most penalized under CE.



Appendix

- Confusion Matrix — Case Study on Lexical Appropriateness

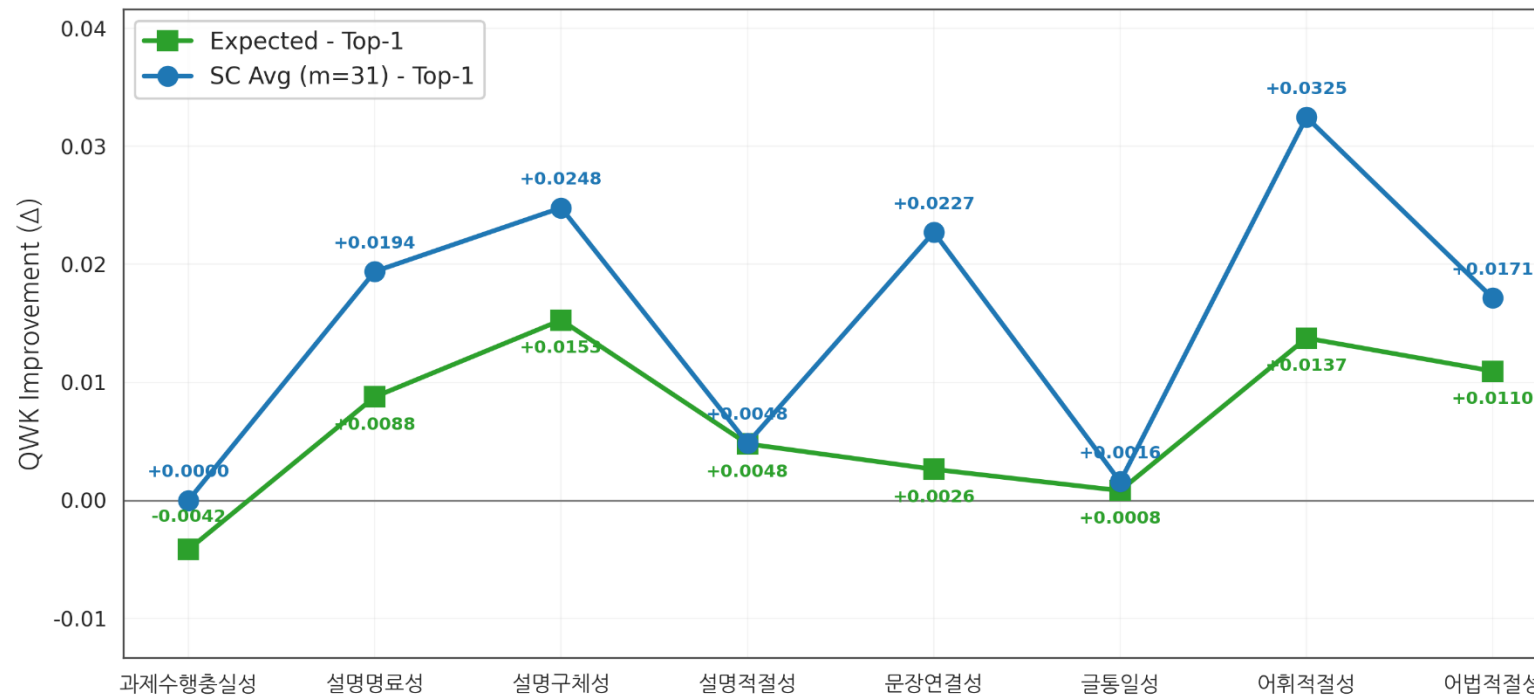
- ❖ Lexical appropriateness has the **lowest inter-rater agreement**, so even human scoring is intrinsically uncertain.
- ❖ The proposed model spreads its errors onto **adjacent labels** rather than collapsing to a single mode — exactly the error structure favored by QWK.



Appendix

- ED vs. SC Improvement by Rubric

- ❖ **ED** provides a lightweight improvement on every rubric without extra generation cost.
- ❖ **SC-Average** delivers the largest gains, especially on subjective rubrics where stochastic sampling is more useful.



Appendix

- Why the Proposed Method Works

- ❖ The improvement is not from one mechanism but from four targeted components — each addressing a separate failure mode of generative AES.

Problem	Proposed Component	Mechanism	Evidence
Ordinal mismatch	WNTL (expected score)	Treats score tokens as ordinal values	pp.23-27
Class imbalance	WNTL (Inverse-frequency weighting)	Up-weights sparse score bands	pp.23-27
Feedback loss dominance	SAL (cosine alignment)	Semantic-level supervision on feedback	pp.23-27
Stochastic generation	ED · SC	Single-pass expectation / consensus aggregation	pp.23-24

<A single LLM can produce **scores** and **feedback** together with improved QWK>